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Declaration of Authorship: *We declare that all materials included in this assessment is the end result of our own work and that due acknowledgement has been given in the bibliography and references to ALL sources be they printed, electronic or personal.*

PITCHINSTINCT

MANAGER SYSTEM

A manager's decision-support tool

User & Technical Documentation

Sports Analytics Tool

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1. Introduction and Purpose

PitchInstinct is a single-page web application that acts as a decision-support system for a manager and their analytics staff. It turns raw match-event data into insights a coaching team actually needs before a fixture: a read on their own squad, a scouting dossier on the next opponent, a recommended starting eleven, forward-looking predictions, and a set of trained machine-learning and spatial models.

The tool is built for the stakeholders of a sporting organisation, primarily the manager and performance analysts, and is organised around a single, real decision: how to prepare for the next match. Every page answers a question a manager would ask in the build-up to a game, and presents the answer visually (broadcast-style cards, ranked bar charts, radars and pitch maps) with a plain-English read-out so the output can be acted on without a statistics background.

1.1 Key features at a glance

Capability	What it gives the manager	Where
Squad intelligence	FIFA-style player cards, in-depth profiles, head-to-head comparisons and form trends	Match Centre
Opponent scouting	An automated dossier: formation, danger men, likely XI, pressing triggers, set pieces and game-state habits	Opponent
Team selection	An opponent-tuned best XI with one-click tactical presets, live swaps and a balance radar	Selection
Predictive models	Three trained, validated models: Expected Goals, match outcome and player form	Intelligence / Predict
Spatial & vision	Pitch control, defensive shape and shot pressure, plus an OpenCV tracking demonstration	Intelligence
Forecasting & in-game	A Monte Carlo season simulation and a live in-match assistant	Predict
Decision read-outs	Every chart carries a plain-English verdict, an 'About this page' note and a glossary	Every page

1.2 Intended stakeholders and decisions

The primary users are the manager and the performance-analysis staff. The tool supports concrete decisions: which eleven to start and in what shape, which opponent player to neutralise, how aggressively to press, when to push or game-manage in-play, and where the team is heading across the season. Secondary users can be sporting directors and coaching staff, they can read the same outputs for recruitment and review.

1.3 What makes it analytical

The application deliberately separates three kinds of analysis so the user always knows how much to trust a number:

- **Descriptive analytics:** metrics computed directly from the event data (pressure resistance, expected threat, passing-network centrality, Elo ratings, set-piece and transition analysis). These describe what happened.
- **Trained machine-learning models:** three genuinely fitted and validated models (Expected Goals, match outcome and player form) that learn patterns and predict. Each is benchmarked against a naive baseline before it is shown.
- **Spatial and computer-vision analytics:** pitch-control, defensive-shape and shot-pressure models built from StatsBomb freeze-frames, plus an OpenCV pipeline that demonstrates how raw broadcast frames become tracking coordinates.

2. Getting Started: Running the App

PitchInstinct is a Streamlit application and runs in the browser from a local server. A marker can run it with the two commands in 2.2 from the project folder. Steps below assume no prior Python experience.

2.1 Requirements

Python 3.9 or newer, plus the packages pinned in requirements.txt: streamlit, statsbombpy, mplsoccer, pandas, numpy, matplotlib, plotly, networkx, pyarrow, scikit-learn, scipy, opencv-python-headless and Pillow. An internet connection is required for the first run only (to download the open data).

2.2 Install and run

1. Install Python 3.9+ from python.org if it is not already installed. On Windows, tick 'Add Python to PATH' during installation.
2. Open a terminal (Command Prompt / PowerShell on Windows, Terminal on macOS) in the project folder, the folder that contains app.py.
3. Install the dependencies:
pip install -r requirements.txt
4. Launch the app:
streamlit run app.py
5. Streamlit prints a local URL (usually http://localhost:8501) and opens it automatically in your default browser. If it does not open, paste that URL into the browser.

```

Terminal 1
(base) sarveshkhade@Sarveshs-MacBook-Air pitchinstinct % streamlit run app.py

You can now view your Streamlit app in your browser.

Local URL: http://localhost:8501
Network URL: http://172.18.254.199:8501

2026-06-30 17:55:53.418 Please replace `use_container_width` with `width`.
`use_container_width` will be removed after 2025-12-31.

For `use_container_width=True`, use `width='stretch'`. For `use_container_width=False`, use `width='content'`.
2026-06-30 17:55:53.437 Please replace `use_container_width` with `width`.
`use_container_width` will be removed after 2025-12-31.

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2026-06-30 17:55:53.462 Please replace `use_container_width` with `width`.
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2026-06-30 17:55:53.474 Please replace `use_container_width` with `width`.
`use_container_width` will be removed after 2025-12-31.

For `use_container_width=True`, use `width='stretch'`. For `use_container_width=False`, use `width='content'`.
2026-06-30 17:55:53.484 Please replace `use_container_width` with `width`.
`use_container_width` will be removed after 2025-12-31.

For `use_container_width=True`, use `width='stretch'`. For `use_container_width=False`, use `width='content'`.

```

2.3 First run and data caching

On first launch the data layer pulls Manchester City's matches, and, lazily, each opponent's full-season matches, from StatsBomb's open data via statsbombpy, then caches the results as Parquet files in a local `.cache/` folder. The first run is therefore slower and needs an internet connection; subsequent runs read from the cache and are fast. Deleting the `.cache/` folder forces a fresh re-fetch.

2.4 Troubleshooting

Symptom	Cause & fix
'streamlit' is not recognised	Python/Streamlit not on PATH. Re-run the pip install, or use 'python -m streamlit run app.py'.
First run is slow or errors on data	It is fetching open data, ensure you are online. Once cached, later runs are fast and offline.
A page is blank or errors after editing	Stop the server (Ctrl+C) and relaunch. Delete <code>.cache/</code> if the data looks stale.
Port 8501 already in use	Run 'streamlit run app.py --server.port 8502' to use another port.

3. Interface and Navigation

The interface is a frosted-glass dark theme over a stadium backdrop, designed to read like a broadcast analysis desk. Navigation and the global match settings live in a fixed bar across the top of every page.

3.1 The top bar

- **Brand and identity:** the PitchInstinct wordmark and club badge on the left.
- **Match settings:** the Opponent and Formation selectors (see Section 4). Changing either re-computes the whole app around the new fixture.
- **Search:** a search icon that expands into a field with live page suggestions, so the user can jump to any page by name.

3.2 Navigation: Grouped Section Menus

Pages are reached from a row of grouped dropdown menus across the top of the page, one per section (Match Centre, Opponent, Selection, Predict, Intelligence, Help). Opening a section reveals its pages; selecting one loads it and highlights it as active. Grouping the pages this way keeps the workflow tidy and mirrors a manager's natural pre-match sequence rather than presenting every page at once.

3.3 Page groups

The pages are grouped into six areas that follow a manager's pre-match workflow: understand your own team, scout the opponent, pick the side, look ahead, dig into the models, and get help:

Group	Pages
Match Centre	Dashboard · Squad · Player Profile · Compare Players · Season Trends · Expected Threat
Opponent	Scout Report · Matchup · Who To Stop · Transition · Tactical Degradation
Selection	Build XI · Availability · Pressing Game · Set Pieces
Predict	Season Forecast · Live Assistant
Intelligence	xG Model · Match Predictor · Form Forecast · Spatial Lab · Vision Lab
Help	How To Use

Every page also carries an 'About this page' explainer and a Glossary dropdown that defines the jargon used on that page in plain English.

4. Configurable User Settings

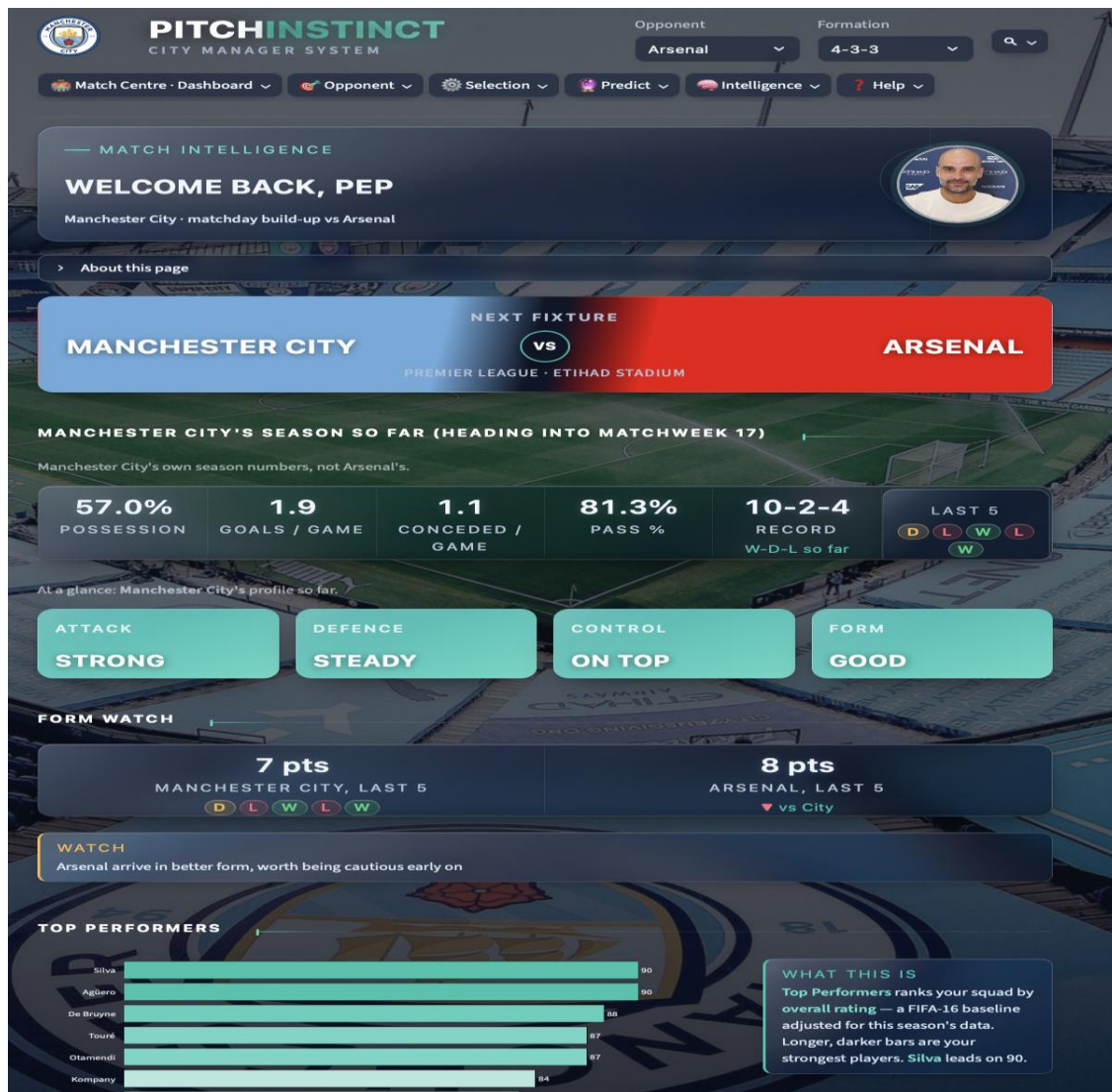
The tool is driven by a small number of global controls plus page-level options. The two global settings below are read by every page, so the entire app re-computes around them.

Setting	Where	Options	Effect
Opponent	Top bar (global)	Any Premier League 2015/16 team	Loads that team's full-season data and drives every Opponent page,

Setting	Where	Options	Effect
			the recommended XI, the projected scoreline and the headline threat.
Formation	Top bar (global)	4-3-3, 4-4-2, 3-5-2	Sets the shape used by the lineup builder and the shape-vs-shape matchup.
Tactical preset	Build XI page	Auto, Tiki-Taka, Gegenpress, Counter, Balanced	Re-weights the best-XI scoring (press resistance / goal threat / passing / pressing) to emulate a managerial style.
Position filter	Squad page	All / GK / DEF / MID / FWD	Filters the squad cards to compare like for like.
Transition window	Transition page	Slider (seconds after a turnover)	Sets how long after winning the ball the opponent's open space is measured.
Score state & minute	Live Assistant	Leading / Level / Trailing, 1 - 90'	Drives the in-game prescriptive calls and the suggested substitution.
Venue	Match Predictor	Home / Away	Feeds the trained outcome model to give Win/Draw/Loss probabilities for the fixture.
Frame upload	Vision Lab	Upload an image (optional)	Runs the computer-vision pipeline on your own frame instead of the built-in synthetic one.

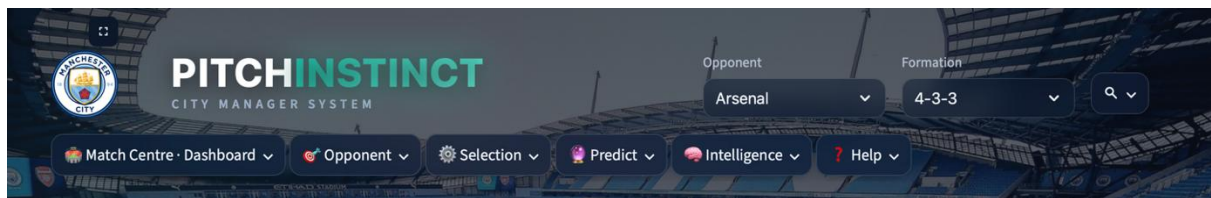
5. Feature Walkthrough

Each page is described below with what it shows, how to use it, and how to interpret the output.



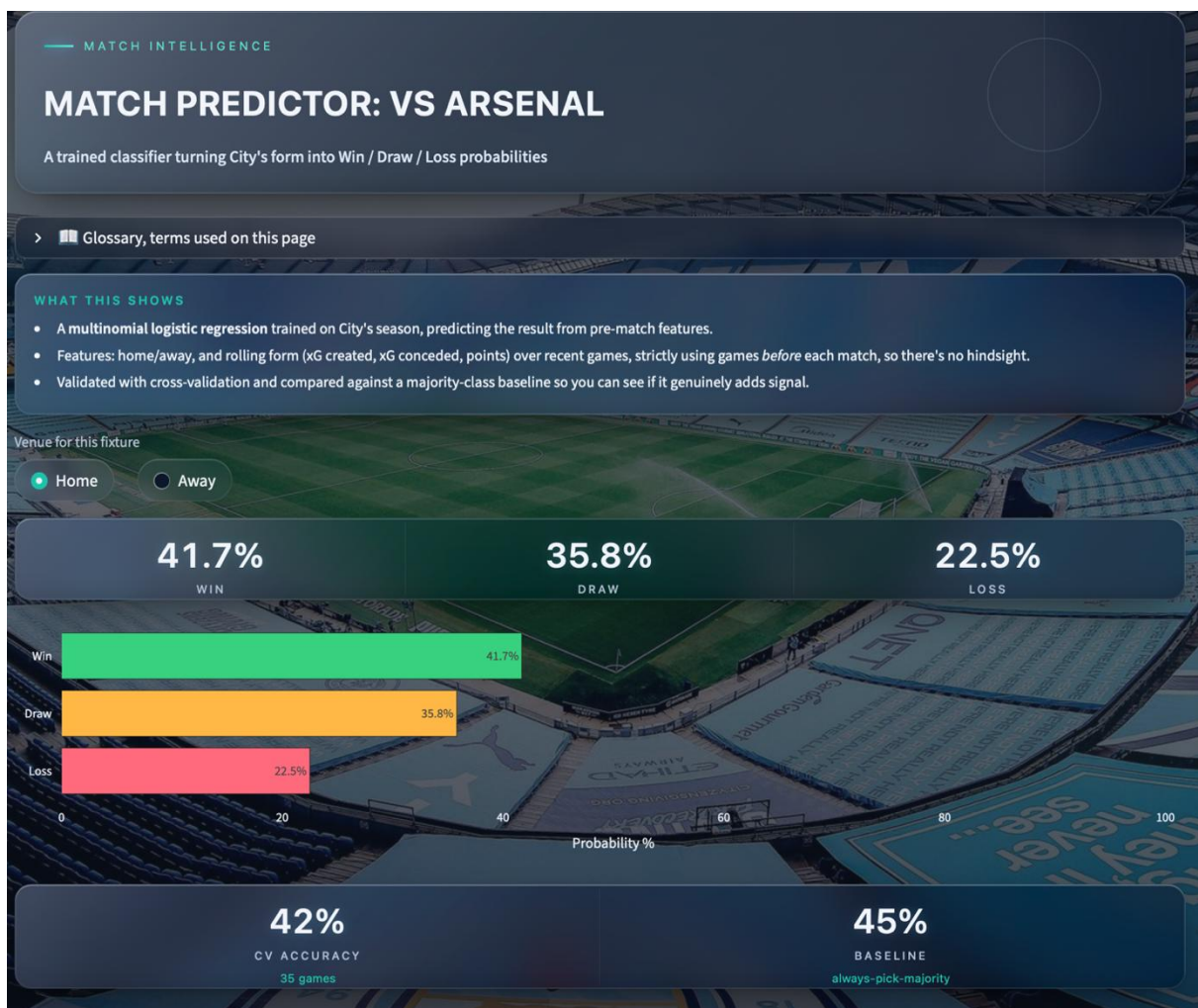
5.1 Match Centre: Know Your Own Team

- **Dashboard:** the pre-match snapshot, covering season form, possession, goals for and against, the recent W-D-L run, an at-a-glance verdict (Attack / Defence / Control / Form), a ranked Top Performers bar chart, the headline threat from the next opponent, and a Season Headlines panel of real, date-aware storylines from the 2015/16 campaign.
- **Squad:** every player as a FIFA-style card (a baseline overall adjusted by season data, plus data-derived attribute bars), filterable by position. Card sizing is fixed so long names do not distort the grid.
- **Player Profile:** one player in depth: attribute bars, season KPIs, a transparent breakdown of how the overall was derived, and a form trend across matchweeks.
- **Compare Players:** two players side by side on a radar and head-to-head, to settle selection calls.
- **Season Trends:** points accumulation and rolling goals for/against to spot momentum and dips; the goals-for line is shown in blue and goals-against in red so the two read apart at a glance.
- **Expected Threat (xT):** who actually moves the ball into dangerous areas, ranked by total threat added, on a threat-value pitch surface.



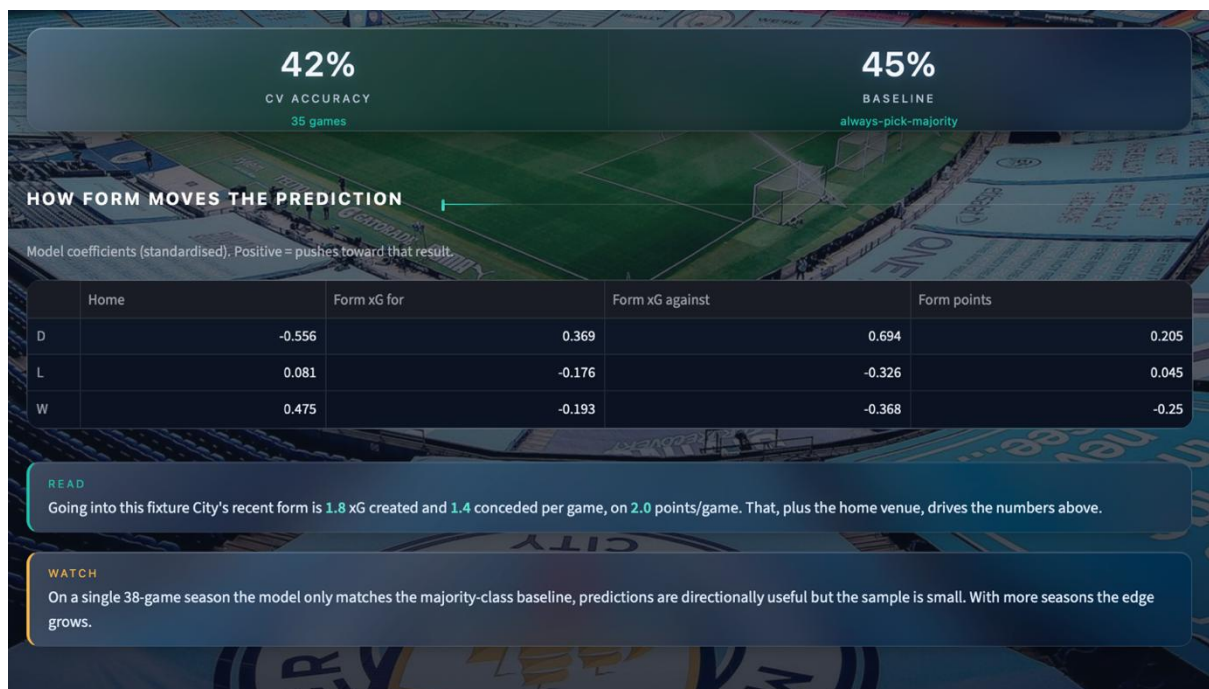
5.2 Opponent: Scout Who You Are Playing

- **Scout Report:** the full dossier across tabs: formation, press intensity, attacking flank, danger men, likely XI, pressing triggers, set-piece profile and game-state tendencies, with pitch maps.
- **Matchup:** your shape versus theirs (City in blue, the opponent in red), the key duels, and a projected scoreline with Win/Draw/Loss odds.
- **Who To Stop:** the opponent's creative hub, found by passing-network betweenness centrality (goalkeepers removed); only the strongest pass combinations are drawn so the network reads cleanly. Neutralise him to break their build-up.
- **Transition:** where space opens up the instant you win the ball back, over a configurable time window.
- **Tactical Degradation:** whether the opponent's shape drops deeper or narrows late in games, and the window to strike.



5.3 Selection: Pick the Team

- **Build XI:** an auto-picked XI tuned to beat the selected opponent, with one-click tactical presets, live player swaps, visible selection scores, a balance radar (recommended vs your pick) and role/familiarity read-outs.
- **Availability:** squad fitness and workload at a glance: knocks, suspensions and minutes/load, built from real injury-substitution, treatment and card events (StatsBomb holds no medical data, so this is inferred honestly from those events plus a workload proxy).
- **Pressing Game:** how hard and how high City press (PPDA), where they press, and the top pressers; also surfaces which of City's own players hold up best when pressed.
- **Set Pieces:** dead-ball delivery zones and set-piece threat, around a third of goals come from here.



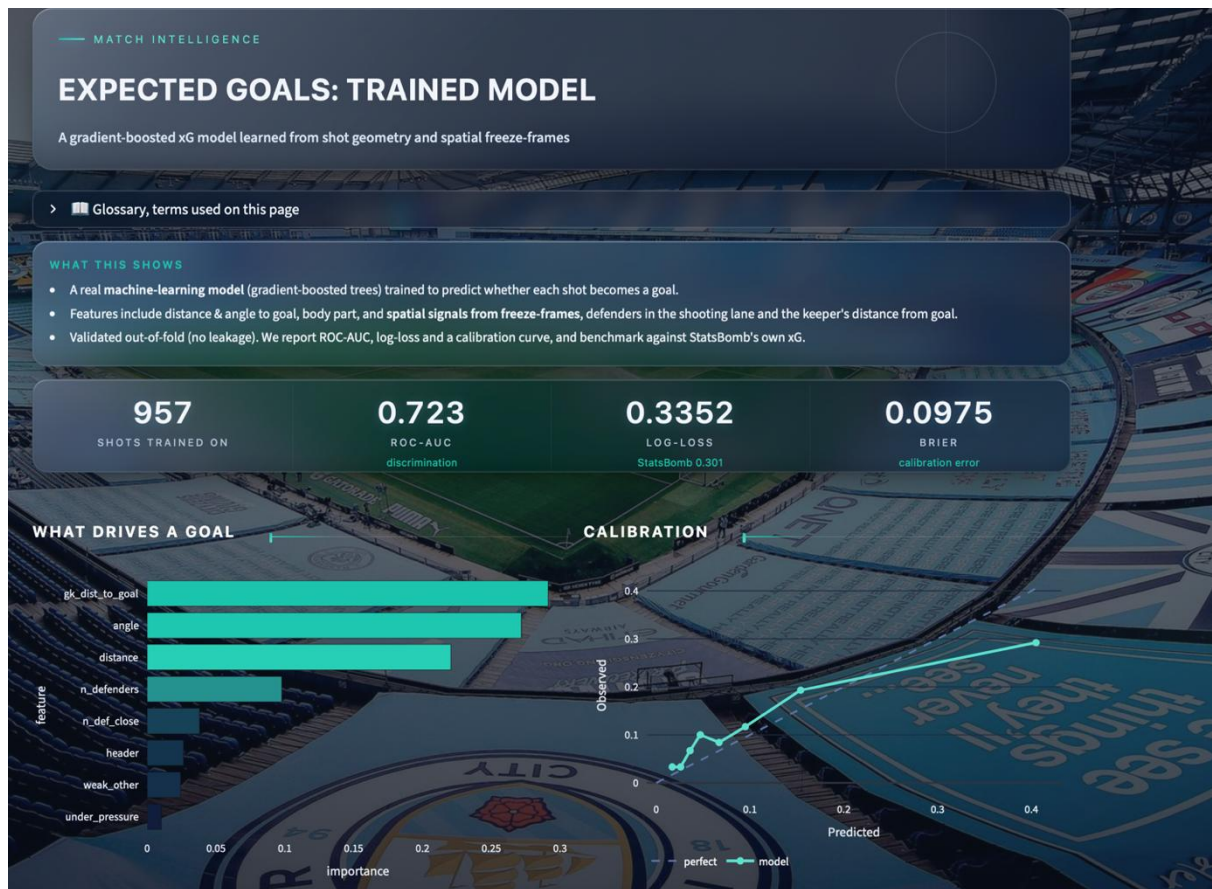
5.4 Predict: Look Ahead

- **Season Forecast:** a Monte Carlo simulation of the season (Elo-rated fixtures played out hundreds of times) giving a predicted finish plus title, top-four and relegation odds, with form projections.
- **Live Assistant:** set the minute and scoreline and the assistant gives the calls to make right now (when to push, game-manage, who to mark) plus a suggested substitution.

5.5 Intelligence: Trained Models, Spatial and Vision

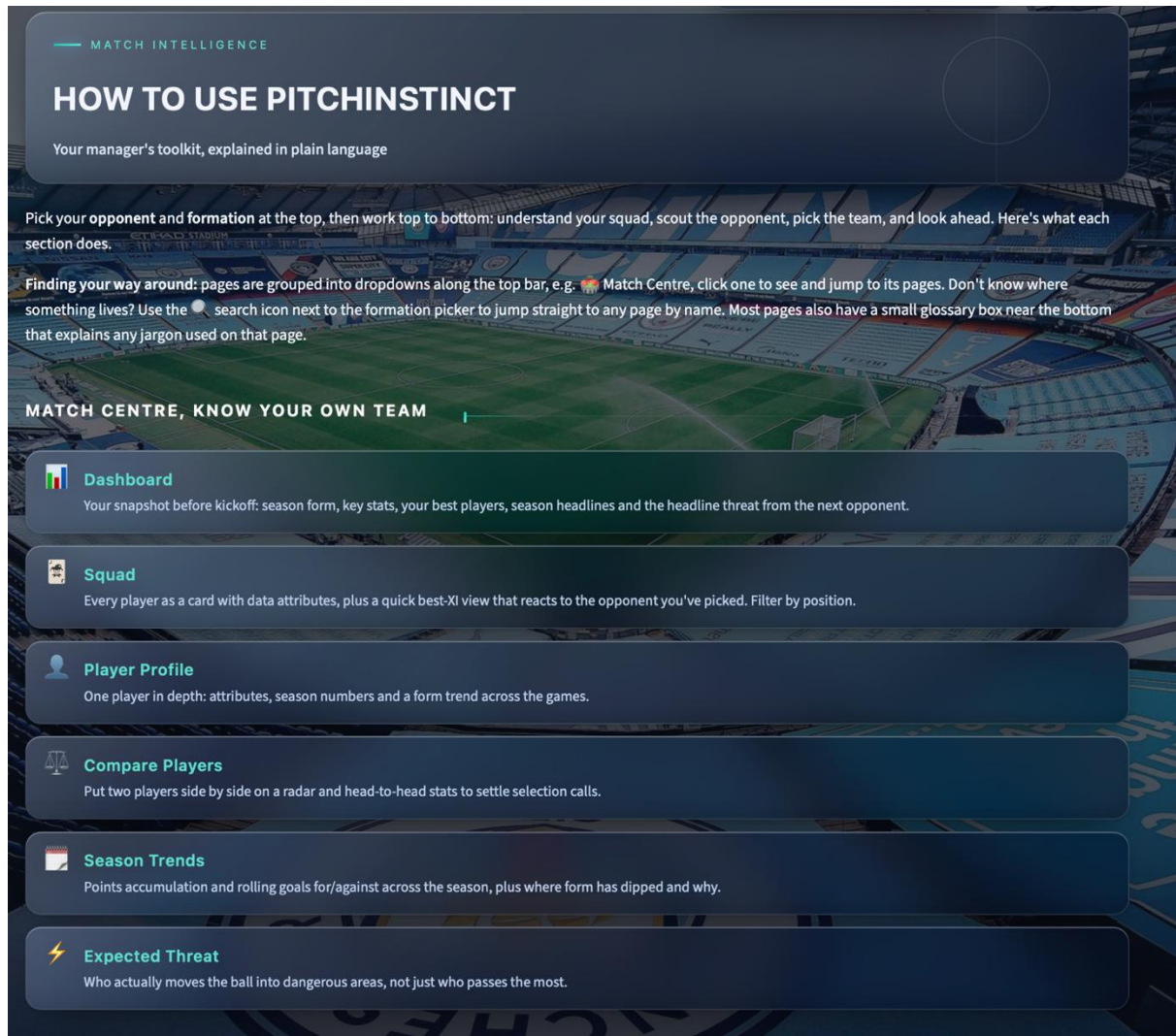
- **xG Model:** a gradient-boosted Expected Goals model trained on shot geometry and spatial freeze-frame features, with ROC-AUC, log-loss, a calibration curve, feature importances and a finishing leaderboard.
- **Match Predictor:** a multinomial logistic model turning rolling form into Win/Draw/Loss probabilities for the next fixture, cross-validated against a majority-class baseline.
- **Form Forecast:** a ridge-regression model projecting each player's next-match goal involvement from rolling form, benchmarked against a predict-the-mean baseline.

- **Spatial Lab:** pitch control, defensive shape and shot pressure computed from per-shot freeze-frame positions (a stand-in for tracking data).
- **Vision Lab:** the computer-vision pipeline that detects players and pitch lines from a frame, assigns teams by colour, and maps pixels onto the pitch, feeding the same spatial model.



5.6 Help

- **How To Use:** a plain-language tour of every section, what each page does and how the pieces fit together.



6. The Analytics Process (Methodology)

This section documents how the numbers are produced, so a stakeholder can understand and trust the outputs.

6.1 Data pipeline

All analysis is built on StatsBomb's free, event-level data for the 2015/16 Premier League (competition_id = 2, season_id = 27). It is the only free season with full event detail, and crucially it covers all twenty teams, which is necessary to scout any opponent's tendencies, not just City's. The data layer fetches and caches City's own events, each opponent's full-season events, squad statistics and player positions. StatsBomb pitch coordinates run $x \in [0,120]$ (own goal to opponent goal) and $y \in [0,80]$.

6.2 Descriptive analytics

- **Pressure resistance:** a player's pass-completion under pressure versus unpressured, the drop between them, and a combined PR score weighted by pitch zone (defensive ×3, middle ×2, attacking ×1).
- **Expected Threat (xT):** a value surface where each pitch zone is worth the goals scored per touch from there; a pass or carry's xT = value(end) – value(start), summed per player.
- **Passing-network centrality:** the opponent's pass network is rebuilt and players ranked by betweenness centrality (networkx) to find the orchestrator to neutralise.
- **Elo ratings:** a self-adjusting league-strength model walked through the season using $P = P^{n-1} + k(w - w_e)$; used for win probabilities and the season simulation, and back-tested against a naive home-win baseline.
- **Transition vulnerability & tactical degradation:** where the opponent's space opens after turnovers, and whether their average defensive position drops late in matches.
- **Set-piece analysis:** delivery zones, dead-ball goals and set-piece threat/weakness.

6.3 Derived FIFA-style ratings

Player overalls are derived, not copied. A FIFA 16 baseline rating is used only as a prior, then adjusted by the player's actual 2015/16 output:

overall = FIFA 16 baseline + adjustment

where adjustment = $0.32 \times \text{delta} \times \text{confidence}$ (capped at ± 5), delta = the player's position-weighted performance minus the average for his position, and confidence = $\min(1, \text{appearances} \div 19)$. Six attribute bars per player are min-max normalised from season KPIs. Goalkeepers are anchored to the baseline because outfield event data cannot fairly rate them. The Player Profile page shows this calculation transparently for each player, so a marker can see the rating is computed rather than copied.

6.4 Trained machine-learning models

Three models are genuinely trained with scikit-learn and validated before being trusted in the UI:

Model	Type	Predicts	Validation
Expected Goals (xG)	Gradient-boosted trees	Probability a shot becomes a goal	Out-of-fold ROC-AUC, log-loss, Brier; calibration curve; benchmarked vs StatsBomb xG
Match outcome	Multinomial logistic regression	Win / Draw / Loss for the next fixture	Cross-validation accuracy vs a majority-class baseline
Player form	Ridge regression	Next-match goal involvement (goals + assists)	Mean-absolute-error vs a predict-the-mean baseline

Features are built strictly from games before each match (rolling windows), so there is no hindsight leakage. Each model is shown only with its metrics, so the user can see whether it adds real signal.

6.5 Spatial and computer-vision analytics

At the instant of every shot, StatsBomb records the position of every visible player. From these freeze-frames the app computes a pitch-control surface (who owns which space, via influence functions), defensive compactness and line height, and a shot-pressure read (defenders in the shooting lane, keeper distance). The Vision Lab demonstrates the upstream pipeline with classical OpenCV (grass masking, line detection, player-blob detection restricted to objects embedded in the grass, team assignment by colour clustering and a homography projection) on a synthetic generated frame, closing the loop from image to coordinates to analytics.

6.6 Forward-looking simulation

The Season Forecast rates all twenty teams with Elo, converts each remaining fixture into expected goals, and plays out the season hundreds of times (Monte Carlo) to produce a distribution of final positions and the title / top-four / relegation odds.

7. Interpreting the Outputs

7.1 Quick reference for the common visuals

Output	How to read it
Verdict cards	A one-word read on Attack / Defence / Control / Form, colour-coded from strong to weak.
Ranked bar charts	Longer, brighter bars are stronger; used for top performers, goal contributions and threat added.
Radars	The larger and more balanced the shape, the more complete the player or team profile.
Pitch heat maps	Brighter zones are where the activity (pressing, threat, space) concentrates.
Passing network	Bigger nodes are more central players; thicker lines are more frequent pass combinations.
Calibration curve	Points on the diagonal mean the model's predicted goal probability matches reality.
Win/Draw/Loss bars	Probabilities for the next fixture; compare against the stated baseline before trusting them.

7.2 Worked Example: Preparing for a Fixture

This scenario shows how a manager moves through the tool, left-to-right across the top bar, to turn raw event data into a team sheet and a game plan. Suppose the next fixture is Manchester City vs Arsenal.

6. **Set the controls.** Choose Arsenal as the Opponent and 4-3-3 as the Formation in the top bar. Every page now re-computes around this fixture.

7. **Read your own state (Dashboard).** The verdict cards summarise City as, for example, an elite attack and a solid defence in good form, and the headline-threat strip names Arsenal's key man.
8. **Scout the opponent (Scout Report).** Confirm Arsenal's shape, press intensity, preferred attacking flank and set-piece threat from the dossier and pitch maps.
9. **Find the man to stop (Who To Stop).** The passing-network centrality ranks Arsenal's orchestrator; assign your nearest midfielder to deny him time.
10. **Pick the team (Build XI).** Start from the opponent-tuned XI, choose a tactical preset (e.g. Gegenpress to press them high), swap any player and watch the balance radar update.
11. **Sense-check the plan (Matchup).** The shape-vs-shape view, key duels and the projected scoreline with Win/Draw/Loss odds tell you where the game is won.
12. **Manage the game (Live Assistant).** On matchday, set the minute and scoreline; at 70' and leading, the assistant gives game-management calls and a substitution.

In a few minutes the manager has gone from raw StatsBomb events to a justified team sheet, a man-marking plan, a pressing approach and an in-game script, each step backed by a visible metric and a plain-English read-out, which is exactly the informed decision-making the tool is designed to support.

8. Technical Architecture

The application is a single Streamlit app (app.py) with a page-routing if/elif chain over a session-state 'page' value. Expensive joins and data loads are cached once at start-up; the three trained models are cached as resources so they are fitted only once per session. Supporting logic is split into focused, pure-function modules:

Module	Responsibility
app.py	Page routing, top-bar navigation/settings, caching, and the per-page UI.
data_loader.py	The only module that talks to statsbombpy; fetches and caches events, matches, squad stats and positions.
analytics.py	Descriptive metrics: pressure resistance, xT, passing-network centrality, Elo, set pieces, best-XI scoring.
features.py	Opponent scouting features, FIFA-style ratings, xT grid, Monte Carlo simulation, availability, live-assistant logic.
scout.py	Builds the opponent profile and recommends City's best XI against it.
predictive.py	The three trained, validated ML models (xG, match outcome, player form).
spatial.py	Freeze-frame analytics: pitch control, defensive shape, shot pressure.
spatial_cv.py	Self-contained OpenCV pipeline on a synthetic pitch frame.
ui.py	All theme/CSS and HTML component builders (cards, heroes, banners, glossary boxes).
player_images.py glossary.py / news.py	/ Player avatars, the in-app glossary, and curated date-aware season storylines.

9. Assumptions and Limitations

- The dataset is the 2015/16 Premier League season; the app is a historical analytical sandbox, and the season-snapshot logic deliberately cuts off at the selected fixture date so results ahead are never revealed.
- Freeze-frames exist only at shots, so spatial analytics use them as a stand-in for full tracking data.
- The computer-vision pipeline runs on a synthetic generated frame (or a user upload) to demonstrate the method; it does not process real broadcast footage.
- StatsBomb holds no medical/injury dataset; the Availability page is inferred from real injury-substitution, treatment and card events plus a workload proxy, not from clinical data.
- On a single 38-game season some models (e.g. match outcome) are directionally useful but sample-limited; the app shows each model's metrics so the user can judge confidence.
- Goalkeeper ratings are anchored to a FIFA baseline because outfield event data cannot fairly rate them.

10. Group Member Contribution Statement

This section records each group member's individual contribution to the project. The work was divided across the main technical workstreams (the data pipeline, the analytics engine, the application and interface, the machine-learning models, and the spatial and computer-vision components), with all members contributing to testing, review and this documentation. The four members contributed equally, at 25% each, and the individual statements below are summarised in the table that follows.

Sarvesh Khade (25214042) led the data layer and the descriptive-analytics engine. He built the `data_loader.py` module that fetches and caches Manchester City's and each opponent's full-season events, squad statistics and player positions from StatsBomb's open data, including the Parquet caching that keeps repeat runs fast and offline. He also wrote the `analytics.py` module, implementing the core descriptive metrics (pressure resistance, Expected Threat, passing-network centrality, Elo ratings, set-piece analysis and the best-XI scoring) that the rest of the application is built on. His estimated contribution is 25%.

Mohammad Bilal (25208446) was responsible for the application structure, user experience and overall presentation. He built `app.py`, including the page-routing logic, the grouped top-bar navigation and global match settings, and the caching that fits the trained models only once per session. He also developed `ui.py` (the frosted-glass theme and the reusable card, hero, banner and glossary components), and led the writing and formatting of this user and technical documentation. His estimated contribution is 25%.

Prabhdeep Singh (25201673) owned the predictive machine-learning work. He built `predictive.py` and trained the three models used in the application: the gradient-boosted Expected Goals model, the multinomial-logistic match-outcome model, and the ridge-regression player-form model. Importantly, he built each model from rolling pre-match features to avoid hindsight leakage and validated all three against naive baselines (out-of-fold ROC-AUC and a calibration curve for xG, cross-validated accuracy

for match outcome, and mean-absolute-error for player form), so that each model is shown only with the metrics that justify trusting it. His estimated contribution is 25%.

Saif Rizvi (25244110) developed the spatial and computer-vision analytics and led the methodology write-up. He built `spatial.py`, which computes the pitch-control surface, defensive shape and shot-pressure reads from StatsBomb freeze-frames, and the self-contained OpenCV pipeline in `spatial_cv.py` that demonstrates grass masking, line and player detection, team assignment by colour and a homography projection from image to pitch coordinates. He also wrote the methodology section documenting how each class of analysis is produced and how far it should be trusted. His estimated contribution is 25%.

Member	Student No.	Key contributions	%
Sarvesh Khade	25214042	Data pipeline & StatsBomb integration (<code>data_loader.py</code>); descriptive-analytics engine (<code>analytics.py</code>)	25%
Mohammad Bilal	25201673	App structure, routing & navigation (<code>app.py</code>); UI theme & components (<code>ui.py</code>); documentation	25%
Prabhdeep Singh	25208446	Trained ML models (xG, match outcome, player form) in <code>predictive.py</code> ; model validation	25%
Saif Rizvi	25244110	Spatial & vision analytics (<code>spatial.py</code> , <code>spatial_cv.py</code>); methodology write-up	25%

All group members confirm that the above is an accurate and agreed record of contributions to the assessment.

11. Glossary of Key Terms

Term	Meaning
xG (Expected Goals)	The probability a shot becomes a goal, from its geometry and context.
xT (Expected Threat)	The increase in goal probability a pass or carry creates by moving the ball to a more dangerous zone.
PPDA	Passes allowed per defensive action; a measure of pressing intensity (lower = more aggressive).
Pressure resistance	How well a player keeps the ball when pressed, versus unpressed.
PR score	A zone-weighted accumulation of a player's retained possessions under pressure.
Betweenness centrality	How much of a team's passing flows through a given player; identifies the orchestrator.
Elo rating	A self-adjusting team-strength number updated after each result.
Pitch control	The share of the pitch a team is favourite to reach first, from player positions.

Term	Meaning
Freeze-frame	StatsBomb's snapshot of every visible player's position at the moment of a shot.
Calibration	Whether a model's predicted probabilities match observed frequencies.
Monte Carlo simulation	Playing the season out many times with random variation to estimate the range of outcomes.
Confidence (rating)	How much a player's data is trusted, scaled by how many games he has played.

Appendix A: Recommended Screenshots

Capturing the following screenshots from the running app will complete the visual documentation; drop them in beside the relevant section.

- Dashboard (Match Centre): hero, verdict cards, Top Performers and the Season Headlines panel.
- Top bar: Opponent and Formation selectors and the grouped section menus.
- Scout Report (Opponent): the dossier tabs and pitch maps.
- Build XI (Selection): the editable lineup, selection scores and balance radar.
- Season Forecast (Predict): the finish distribution and odds.
- xG Model (Intelligence): calibration curve and feature importances.
- Vision Lab: the detected frame and its mapping onto the pitch.